Module 5 : ITIL Management Practices

1. Core IT Management

1.1 Architecture Management

Architecture Management ensures that the IT systems, applications, data, and infrastructure are designed in a structured way so they support the organization's business goals.

It defines how different IT components work together.

Objectives:

- **Align IT systems with business strategy**
- **Ensure systems are scalable and secure**
- **Reduce duplication of technology**
- **Improve efficiency of IT services**

Components:

1. **Business Architecture – Business processes and strategy**
2. **Application Architecture – Software applications used**
3. **Data Architecture – Data structure and storage**
4. **Technology Architecture – Hardware, servers, networks**

Real-Life Example:

A large bank like HDFC Bank manages thousands of services such as:

- **Mobile banking**
- **ATM systems**
- **Online payments**

Architecture management ensures all these systems connect securely and share data correctly.

For example:

- **Mobile banking app connects with core banking servers**
- **Fraud detection systems analyze transactions in real time**

Without architecture management, these systems would be disconnected and inefficient.

1.2 Knowledge Management

Knowledge Management ensures that important knowledge is collected, stored, shared, and used effectively within the organization.

Types of Knowledge

1. **Explicit Knowledge**
    - **Documented information**
    - **Manuals, reports, guides**

2. **Tacit Knowledge**
    - **Experience of employees**
    - **Skills and problem-solving abilities**

Goals

- **Improve decision making**
- **Avoid repeating mistakes**
- **Help employees solve issues faster**

Tools

- **Knowledge databases**
- **Wikis**
- **Documentation portals**
- **FAQs**

Real-Life Example

Companies like Amazon store solutions to technical problems in internal knowledge bases.

If a server fails:

- **Engineers search the knowledge base**
- **Find previous solutions**

- **Fix the issue quickly**

This reduces downtime and saves time.

1.3 Measurement and Reporting

This practice focuses on collecting and analyzing data to measure IT service performance.

Organizations track Key Performance Indicators (KPIs).

Objectives

- **Monitor service performance**
- **Identify problems**
- **Improve decision making**

Common Metrics

- **System uptime**
- **Incident resolution time**
- **Customer satisfaction**
- **Service availability**

Types of Reports

1. **Operational Reports**

2. Performance Reports
3. Strategic Reports

Real-Life Example

Streaming services like Netflix measure:

- Server uptime
- Video streaming speed
- User buffering rate

If buffering increases, engineers investigate and improve servers.

1.4 Continual Improvement

Continual improvement focuses on constantly improving services, processes, and performance.

ITIL uses the Continual Improvement Model.

Steps

1. What is the vision?
2. Where are we now?
3. Where do we want to be?
4. How do we get there?
5. Take action

6. **Did we achieve results?**
7. **Maintain momentum**

Benefits

- **Better service quality**
- **Lower costs**
- **Higher customer satisfaction**

Real-Life Example

Food delivery platform Swiggy constantly improves its app.

Improvements include:

- **Faster delivery tracking**
- **Better search for restaurants**
- **AI-based recommendations**

This is an example of continual service improvement.

---

2. Governance and Security

2.1 Information Security Management

Meaning

Information Security Management ensures that organizational data is protected from unauthorized access, misuse, or loss.

Security Principles

1. **Confidentiality**
    - **Data accessible only to authorized users**
2. **Integrity**
    - **Data remains accurate and unchanged**
3. **Availability**
    - **Data available when needed**

Security Measures

- **Encryption**
- **Firewalls**
- **Authentication systems**
- **Access controls**

Real-Life Example

Payment company PayPal protects customer data using:

- **Encryption**
- **Multi-factor authentication**

- **Fraud detection systems**

This prevents hackers from stealing financial information.

2.2 Risk Management

Meaning

Risk management identifies potential threats that could affect IT services and develops plans to minimize their impact.

Types of Risks

- **Cyber attacks**
- **System failures**
- **Data loss**
- **Natural disasters**

Risk Management Steps

1. **Identify risks**
2. **Analyze risks**
3. **Evaluate impact**
4. **Plan mitigation**
5. **Monitor risks**

Real-Life Example

Cloud providers like Microsoft manage risks by:

- **Maintaining backup data centers**
- **Disaster recovery systems**
- **Security monitoring**

If one server fails, another server takes over immediately.

2.3 Organizational Change Management

Meaning

Organizational Change Management helps employees adapt to new technologies, processes, or organizational changes.

Goals

- **Reduce resistance to change**
- **Help employees learn new systems**
- **Ensure smooth transitions**

Key Activities

- **Communication**
- **Training**

- **Stakeholder engagement**
- **Support programs**

Real-Life Example

When companies adopt cloud systems like Microsoft Azure, employees must learn new tools.

The organization provides:

- **Training programs**
- **Workshops**
- **Support teams**

This helps employees adapt to the new technology smoothly.

---

3. Strategic IT Planning

3.1 Portfolio Management

Meaning

Portfolio Management manages all IT projects and services to ensure they align with business strategy.

Objectives

- **Prioritize important projects**
- **Allocate resources efficiently**
- **Balance risk and benefits**

Example Portfolio

A telecom company portfolio may include:

- **5G network expansion**
- **Customer mobile app**
- **AI chatbots**
- **Cybersecurity upgrades**

Real-Life Example

Telecom company Reliance Jio manages projects like:

- **5G deployment**
- **Fiber internet rollout**
- **Streaming services**

Portfolio management ensures resources are allocated to high-value projects first.

3.2 Project Management

Meaning

Project Management involves planning, executing, and controlling projects to achieve specific goals within time and budget.

Project Phases

1. **Initiation**
2. **Planning**
3. **Execution**
4. **Monitoring**
5. **Closing**

Example

Developing a mobile banking app involves:

- **Planning features**
- **Designing UI**
- **Coding**
- **Testing**
- **Launching**

Real-Life Example

Companies like Infosys manage large IT projects for global clients using structured project management frameworks.

3.3 Strategy Management

Meaning

Strategy management ensures IT strategy supports overall business goals.

Key Activities

- **Analyzing business environment**
- **Identifying opportunities**
- **Creating IT strategies**
- **Monitoring results**

Strategic Tools

- **SWOT analysis**
- **PESTLE analysis**
- **Balanced scorecard**

Real-Life Example

Tech company Apple aligns IT strategy with business strategy.

Example:

- **Integration of hardware + software ecosystem**
- **Development of secure operating systems**
- **Cloud services integration**

---

4. Self-Study Topics

4.1 Service Financial Management

Manages budgeting, accounting, and cost control for IT services.

Example:
Companies like Google analyze costs of cloud infrastructure to optimize spending.

4.2 Supplier Management

Ensures external vendors deliver quality services and products.

Example:
Companies depend on suppliers like Cisco for networking equipment.

Supplier management ensures:

- **Quality products**
- **Reliable delivery**
- **Contract compliance**

4.3 Workforce and Talent Management

Ensures the organization has skilled employees to deliver IT services.

Includes:

- **Recruitment**
- **Training**
- **Performance management**

Example:
Tech companies like Tata Consultancy Services run large training programs for software engineers.

4.4 Relationship Management

Manages relationships with customers and stakeholders.

Goals:

- **Understand customer needs**
- **Improve satisfaction**
- **Build long-term partnerships**

Example:
Companies like Salesforce use CRM systems to maintain strong relationships with clients.

Quick Exam Summary Table

Practice	Purpose
Architecture Management	Design IT systems structure
Knowledge Management	Store and share knowledge
Measurement & Reporting	Monitor IT performance
Continual Improvement	Improve services continuously
Information Security	Protect data
Risk Management	Identify and reduce risks
Organizational Change	Manage employee adaptation

Portfolio Management	**Manage all IT investments**
Project Management	**Execute IT projects**
Strategy Management	**Align IT with business goals**

Date 20/02/2026

4. THE ITIL SERVICE VALUE SYSTEM

The ITIL SVS describes how all the components and activities of the organization work together as a system to enable value creation.

Each organization's SVS has interfaces with other organizations, forming an ecosystem that can in turn facilitate value for those organizations, their customers, and other stakeholders

The ITIL SVS ensures that:

- Customer **needs (demand)**
  ⬇
- Go through structured activities
  ⬇
- To produce **value (useful services)**

It focuses on **value co-creation** between service provider and customer.

🔶 Components of ITIL Service Value System

There are **5 key components**:

1. **Guiding Principles**
2. **Governance**
3. **Service Value Chain**
4. **Practices**
5. **Continual Improvement**

1️⃣ Guiding Principles

These are universal recommendations that guide organizations in all circumstances.

- Focus on value
- Start where you are
- Progress iteratively with feedback
- Collaborate and promote visibility
- Think and work holistically
- Keep it simple and practical
- Optimize and automate

ITIL 4 Guiding Principles:

1️⃣ Focus on Value

👉 Always ensure the activity delivers value to customers or stakeholders.

✅ Example:

An online banking app plans to add a new theme color.
Instead, they prioritize **instant fund transfer improvement**, because customers care more about faster transactions than appearance.

✔ Value delivered = Faster service for customers.

2️⃣ Start Where You Are

👉 Do not build everything from scratch. Assess current state and reuse what works.

✅ Example:

A company wants a new IT service desk tool.
Before purchasing new software, they review their existing ticketing system and improve its configuration instead of replacing it.

✔ Saves cost and time.

3️⃣ Progress Iteratively with Feedback

👉 Implement changes in small steps and gather feedback.

✅ Example:

An e-commerce company launches a chatbot:

- Phase 1: Basic FAQs
- Phase 2: Order tracking

- Phase 3: Refund processing

After each release, they collect user feedback and improve it.

✔ Reduces risk and improves quality.

4️⃣ Collaborate and Promote Visibility

👉 Work together and make work transparent.

✅ Example:

During a major system upgrade:

- IT team
- Security team
- Business team

All collaborate using shared dashboards showing progress and risks.

✔ Avoids miscommunication and delays.

5️⃣ Think and Work Holistically

👉 Consider the whole organization, not just one department.

✅ Example:

Before changing a payment gateway, the company checks impact on:

- Customers
- Finance department

- Support team
- Security team

✔ Ensures smooth integration without breaking other processes.

6️⃣ Keep It Simple and Practical

👉 Avoid unnecessary complexity.

✅ Example:

Instead of creating a 20-step approval process for minor IT changes, the company creates:

- Simple approval for low-risk changes
- Detailed approval only for high-risk changes

✔ Saves time and improves efficiency.

7️⃣ Optimize and Automate

👉 Improve processes first, then automate.

✅ Example:

An IT team notices many password reset requests.

Step 1: Simplify reset process
Step 2: Implement self-service password reset portal

✔ Reduces service desk workload.

Example: 1

A company implementing a **ticketing system**:

- Instead of buying a new tool immediately, they review their current process first (**Start where you are**).
- They release features in phases (**Progress iteratively**).
- They automate password resets (**Optimize and automate**).

Example: 2

A food delivery company:

- Improves app speed instead of building a new app (**Start where you are**).
- Automates order confirmation messages (**Optimize and automate**).
- Collects user feedback before launching new features (**Progress iteratively**).

---

2 Governance

Governance ensures the organization's direction and compliance.

It answers:

- Are we doing the right things?
- Are we meeting stakeholder expectations?

Example: 1

A bank's IT department:

- Ensures IT projects align with business goals.
- Approves budget for cybersecurity upgrades.
- Reviews compliance with data protection laws.

Senior management evaluates performance and ensures accountability.

Example: 2

Company management:

- Approves budget for cloud upgrade.
- Ensures payment systems follow security standards.
- Reviews service performance reports monthly.

---

3 Service Value Chain (Core of SVS)

The **Service Value Chain** is the operating model of ITIL.

It has **6 activities**:

1. **Plan**
2. **Improve**
3. **Engage**
4. **Design & Transition**
5. **Obtain/Build**
6. **Deliver & Support**

These activities combine to create value streams.

1️⃣ Plan

👉 Align IT strategy with business goals.

This activity ensures everyone understands:

- Vision
- Current status
- Future direction

✅ Example:

The company decides to introduce **Live Classes** to compete with other platforms.
Management defines:

- Budget
- Timeline
- Technology requirements

✔ Goal: Increase student engagement by 30%.

2️⃣ Improve

👉 Continuously improve products, services, and practices.

Improvement happens at all levels.

✅ Example:

The company reviews feedback from recorded classes.
Students request real-time doubt clarification.
So, the improvement idea: **Add live interaction feature**.

✔ Service enhancement based on feedback.

3️⃣ Engage

👉 Interact with stakeholders (customers, users, suppliers).

This ensures good communication and understanding of needs.

✅ Example:

The company:

- Surveys students about live class timings.
- Talks with trainers about technical requirements.
- Coordinates with cloud providers.

✔ Ensures stakeholder expectations are understood.

4️⃣ Design & Transition

👉 Design the service and move it into production safely.

Includes:

- Testing
- Risk assessment
- Deployment planning

✅ Example:

IT team:

- Designs live streaming architecture.

- Tests video quality and security.
- Conducts pilot launch with 100 students.

✔ Ensures smooth transition to live environment.

5️⃣ Obtain/Build

👉 Acquire or develop necessary components.

This may involve:

- Buying software
- Developing code
- Setting up infrastructure

✅ Example:

Developers:

- Build live streaming module.
- Integrate chat feature.
- Purchase additional cloud bandwidth.

✔ Service components are created.

6️⃣ Deliver & Support

👉 Deliver the service and provide ongoing support.

Includes:

- User support

- Incident management
- Monitoring

✅ Example:

After launch:

- Students attend live classes.
- Helpdesk resolves login issues.
- IT monitors server performance.

✔ Service is actively delivering value.

🔄 How It All Connects

Customer Demand
⬇
Engage (Understand needs)
⬇
Plan (Set strategy)
⬇
Design & Build
⬇
Deliver & Support
⬇
Improve (Based on feedback)
⬇
More Value Created

Activity	Purpose	Example

Plan	Set direction	Define live class strategy
Improve	Enhance service	Add interaction feature
Engage	Communicate	Survey students
Design & Transition	Prepare service	Test live streaming
Obtain/Build	Develop components	Build streaming module
Deliver & Support	Run & support service	Provide helpdesk support

---

Example 1 – Mobile Banking App

Let’s take a banking mobile app launch:

Activity	Real-Time Example
Plan	Define digital banking strategy
Engage	Collect customer feedback

Design & Transition	Design secure UI & test
Obtain/Build	Developers build app
Deliver & Support	Release app & provide helpdesk
Improve	Add new features based on feedback

This cycle continues.

Example 2 – Adding “Live Order Tracking”

Activity	Real Example
Plan	Decide to improve customer tracking
Engage	Collect customer feedback
Design & Transition	Design GPS tracking feature
Obtain/Build	Developers build feature
Deliver & Support	Launch feature & support users
Improve	Enhance tracking accuracy

This flow converts **customer demand** → **valuable feature**.

4️⃣ ITIL Practices

Practices are sets of organizational resources designed for performing work.

They are grouped into:

🟦 General Management Practices

Example:

- Continual Improvement
- Risk Management
- Project Management

🟩 Service Management Practices

Example:

- Incident Management
- Problem Management
- Change Enablement
- Service Desk

🟥 Technical Management Practices

Example:

- Deployment Management
- Infrastructure Management

Example 1 – Incident Management

If a company's email server crashes:

- User raises ticket.
- Service desk logs incident.
- IT team restores service.
- Problem team investigates root cause.

This uses **Incident + Problem + Change Enablement practices** together.

Example: 2

If the app crashes:

- **Incident Management** → Restore service quickly
- **Problem Management** → Find root cause
- **Change Enablement** → Approve fix deployment

---

5 Continual Improvement

Continuous alignment of services with business needs.

It follows the **Continual Improvement Model**:

1. What is the vision?
2. Where are we now?
3. Where do we want to be?
4. How do we get there?

5. Take action
6. Did we get there?
7. How do we keep momentum?

Example 1

An e-commerce company:

- Measures website downtime.
- Finds frequent payment failures.
- Improves server capacity.
- Monitors improvement metrics.

Example: 2

Company notices:

- Delivery delays during peak hours
  Solution:
- Hire more delivery partners
- Upgrade server capacity
- Monitor improvements

🔷 How Everything Works Together

The SVS shows:

Stakeholder Demand
⬇
Service Value Chain Activities
⬇
Practices support activities
⬇
Governance directs
⬇
Continual Improvement enhances
⬇
Value Co-Creation

Real-Life Example (Full Scenario)

Company: Online Shopping Platform

Problem: Website slow during sales

Using SVS:

- Governance approves upgrade budget
- Guiding principle: Optimize & automate
- Service value chain: Plan → Build → Deliver
- Practices: Incident, Change, Capacity management
- Continual improvement: Monitor performance metrics

Result: Faster website + better customer experience = Value creation

End

Simple ITIL Stakeholder Relationship Example

ITIL framework, stakeholders are individuals or groups that have an interest in IT services, their management, and outcomes.

For an **online banking service**:

- Customer → Bank Management
- Sponsor → CFO
- Users → Bank staff & customers
- Service Provider → IT Department
- Supplier → Cloud vendor
- Regulator → Banking authority

1️⃣ Product

A **product** is a configuration of resources (people, technology, processes) created by an organization and offered to customers.

👉 It is something the provider creates.

🔹 **Example:**

- A **Laptop** sold by Dell
- A **Banking software application**
- A **Cloud storage platform**

In IT:
Microsoft Office 365 is a **product** created by Microsoft.

2️⃣ Service

A **service** is a way of delivering value to customers by facilitating outcomes they want, **without the customer managing specific costs and risks**.

👉 Customer gets the result, but doesn't worry about how it works.

🔹 **Example:**

- Gmail service → You send emails without managing servers.

- Uber → You travel without owning a car.
- Cloud hosting → You run applications without managing data centers.

In IT:
A company using AWS cloud doesn't manage physical servers — AWS handles it. That is a **service**.

3️⃣ Product Offering

A **product offering** is a specific version or package of a product made available to customers.

👉 A product can have multiple offerings.

🔹 Example:

Microsoft 365 has:

- Basic Plan
- Business Standard
- Enterprise Plan

Each plan is a **product offering**.

Another example:
A mobile phone model (product)
128GB version and 256GB version → different **product offerings**.

4️⃣ Service Relationship

A **service relationship** is the cooperation between service provider and service consumer.

It includes:

- Service provision (provider delivers)
- Service consumption (customer uses)
- Service relationship management (communication & improvement)

🔹 **Example:**

An IT company providing AMC (Annual Maintenance Contract) to a bank:

- IT company provides support
- Bank uses the service
- Both communicate for issue resolution

That ongoing interaction is a **service relationship**.

VALUE CONCEPTS

ITIL says value depends on:

5️⃣ Value = Outcomes – Costs – Risks

Value is created when:

- Desired outcomes are achieved
- Costs are acceptable
- Risks are minimized

🔹 **Example: Cloud Backup Service**

Company wants:

- Secure data backup (Outcome)
- Affordable pricing (Cost)
- No data loss risk (Risk)

If:
✔ Data is safe
✔ Cost is reasonable
✔ Risk of data breach is low

👉 High value is delivered.

If:
❌ Cost is too high
❌ Frequent data loss

👉 Value decreases.

6️⃣ Outcomes

Outcomes are the **results customers want**.

👉 Focus on results, not activities.

🔹 **Example:**

- Bank wants secure transactions.
- Student wants to pass exam.
- Company wants 99.9% system uptime.

These are **outcomes**.

7️⃣ Costs

Costs are the money, time, and resources spent.

🔹 **Example:**

- Subscription fees
- Maintenance cost
- Employee training cost

If cost is too high compared to benefit → value reduces.

8️⃣ Risks

Risk = possibility of something going wrong.

🔹 **Example:**

- Server downtime
- Cyber attack
- Data loss
- Compliance penalties

Good services reduce customer risk.

UTILITY AND WARRANTY

These are very important ITIL concepts.

9️⃣ Utility (Fit for Purpose)

Utility means **what the service does**.

👉 Does it meet functional requirements?

🔹 Example:

A printer:

- Can print documents
- Can scan

That is its **utility**.

Cloud storage:

- Stores data
- Allows sharing

That is its **utility**.

Simple meaning:
👉 "What the service does"

1️⃣0️⃣ Warranty (Fit for Use)

Warranty means **how well the service performs**.

It includes:

- Availability
- Capacity
- Continuity
- Security

🔹 Example:

Cloud service:

- 99.9% uptime
- Fast response time
- Secure access
- Backup available

That is **warranty**.

Simple meaning:
👉 "How well it works"

🔥 Easy Combined Example

Example: Online Banking App

Utility:

✔ Transfer money
✔ Check balance
✔ Pay bills

(What it does)

Warranty:

✔ Available 24/7
✔ Secure login
✔ Fast transaction
✔ No crashes

(How well it performs)

If app works but crashes daily → Utility is good, Warranty is poor.

Quick Comparison Table

Term	Simple Meaning	Example
Product	Something created	Laptop
Service	Delivering outcome without customer managing risk	Cloud hosting
Product Offering	Version/package of product	Basic / Premium plan
Outcome	Result customer wants	Secure transactions
Cost	Money/time spent	Subscription fee
Risk	What can go wrong	Data breach
Utility	What service does	Send email
Warranty	How well it works	99.9% uptime

Four Dimensions of Service Management

In **ITIL 4**, the **Four Dimensions of Service Management** ensure that a service is designed and managed in a balanced way.
If even one dimension is weak, the service may fail.

The four dimensions are:

1. **Organizations & People**
2. **Information & Technology**
3. **Partners & Suppliers**
4. **Value Streams & Processes**

Let's explain each with **real-life ITSM examples**.

---

1 Organizations & People

This dimension focuses on:

- Roles & responsibilities
- Skills & competencies

- Culture & communication
- Organizational structure

👉 Even the best tools fail if people are not skilled or aligned.

🔹 Real-Life ITSM Example:

A company implements **ServiceNow** for IT ticketing.

But:

- Employees don't know how to log tickets.
- IT support staff are not trained.
- No clear incident escalation matrix.

Result → Delays, confusion, unhappy users.

When fixed:

- Training is provided.
- Roles defined (L1, L2, L3 support).
- SLA responsibilities assigned.

Now service improves.

✅ This is the **Organizations & People** dimension in action.

2️⃣ Information & Technology

This dimension includes:

- Applications

- Databases
- Infrastructure
- Tools
- Security
- Information flow

👉 Technology must support service delivery.

🔹 Real-Life ITSM Example:

An IT company provides **24/7 monitoring service**.

They need:

- Monitoring tool (e.g., SolarWinds)
- Ticketing system
- CMDB
- Secure network
- Backup system

If monitoring tool fails → incidents are not detected.

If database crashes → ticket history lost.

Technology supports service delivery → this is **Information & Technology**.

3️⃣ Partners & Suppliers

Most organizations depend on external vendors.

This dimension covers:

- Cloud providers
- Hardware vendors
- Software vendors
- Outsourcing partners
- Contracts & SLAs

🔹 Real-Life ITSM Example:

A bank uses:

- AWS for cloud hosting
- Cisco for networking
- TCS for IT support
- Microsoft for licenses

If AWS has outage → banking app goes down.

Even though bank IT team is strong, they depend on **partners**.

Managing vendor contracts, SLAs, and performance is part of this dimension.

4️⃣ Value Streams & Processes

This dimension focuses on:

- Workflows
- Activities

- Procedures
- End-to-end service delivery

👉 How work moves from request to resolution.

🔹 Real-Life ITSM Example:

Employee laptop issue:

Step 1 → User raises ticket
Step 2 → L1 support checks
Step 3 → If hardware issue → escalated to L2
Step 4 → Replacement approved
Step 5 → Issue closed

This structured flow is a **Value Stream**.

Processes involved:

- Incident Management
- Change Management
- Asset Management

If process is unclear → delays happen.

🔥 Combined Example (Complete ITSM Scenario)

Let's take: **Company providing Email Service**

1. Organizations & People

- IT admins

- Helpdesk team
- Security team

2. Information & Technology

- Email servers
- Spam filters
- Backup systems
- Authentication tools

3. Partners & Suppliers

- Microsoft (Exchange Online)
- Internet provider
- Security vendor

4. Value Streams & Processes

- New mailbox creation process
- Password reset process
- Incident resolution process

If all four dimensions work properly → Email service runs smoothly.

🎯 Why Four Dimensions Are Important

Imagine this situation:

Dimension Weak	What Happens
People weak	Poor support
Technology weak	Frequent outages
Partners weak	SLA breaches
Process weak	Slow resolution

All four must be balanced.

📝 Simple Exam Definition (Short Format)

The Four Dimensions of Service Management ensure holistic service management by considering:

1. Organizations & People – Human and structural aspects
2. Information & Technology – Tools and systems
3. Partners & Suppliers – External dependencies
4. Value Streams & Processes – Workflows and procedures

They ensure effective value creation.